Generative AI in Undergraduate Education: Impacts on Critical Thinking
Authors: Smith, J. R., & Lee, H. J. (2023)
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ABSTRACT:
This empirical study examines the relationship between generative AI reliance
and critical thinking disposition among university students over one semester.
Sample size: N = 315 undergraduate students assessed via Cornell Critical Thinking Test.

KEY EMPIRICAL FINDINGS:
1. Passive Dependency Risk: Students who copied AI responses without critical review
   showed a 22.4% decline in analytical reasoning and argument evaluation scores ($p = 0.002$).
2. Active Verification Benefit: Students who cross-checked AI outputs against primary
   academic sources improved critical evaluation scores by 14.2%.
3. Threshold Effect: Over-reliance (>12 hours/week of unguided AI usage) correlated
   negatively with independent problem-solving retention ($r = -0.31$).